

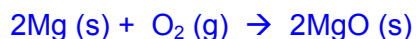
Section A

Answer **ALL** questions in this section on the space provided.

- 1 (a) This part of the question is about the chemistry of Period 3 elements and their compounds.

- (i) Describe what you see when magnesium burns in oxygen and write the equation for the reaction described.

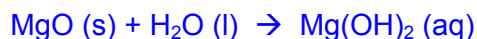
Mg burns brightly with white flame in oxygen to give MgO.



- (ii) Describe what you see when magnesium oxide reacts with water containing Universal Indicator. Write the equation for the reaction described.

MgO (s) is an ionic and basic oxide. It reacts very slowly with water to form weakly alkaline solution of Mg(OH)₂ (aq).

Thus, universal indicator gives a bluish green/blue/greenish blue colour.



- (iii) Account for the difference in structure and bonding between magnesium and silicon.

Mg has giant metallic structure with strong electrostatic forces of attraction between delocalised electrons and Mg ions.

Si has giant covalent structure with covalent bonds between Si atoms.

- (b) The rate of reaction between 2-bromo-2-methylpropane, (CH₃)₃CBr, and aqueous sodium hydroxide when heated under reflux is found to be independent of [OH⁻] but directly proportional to [(CH₃)₃CBr]. Hence the rate equation is

$$\text{Rate} = k[(\text{CH}_3)_3\text{CBr}]$$

- (i) Given that the rate constant for the reaction is $1.09 \times 10^{-3} \text{ s}^{-1}$, calculate the time taken for (CH₃)₃CBr to decrease to 6.25% of its original concentration.

$$t_{1/2} = \ln 2 / (1.09 \times 10^{-3}) = 635.9 \text{ s}$$

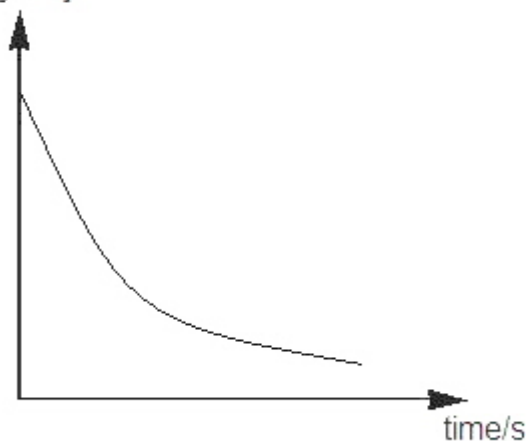
4 half-lives have passed.

$$\text{time taken} = 635.9 \times 4 = 2544 = 2540 \text{ s}$$

- (ii) Sketch the graph of concentration of (CH₃)₃CBr versus time.

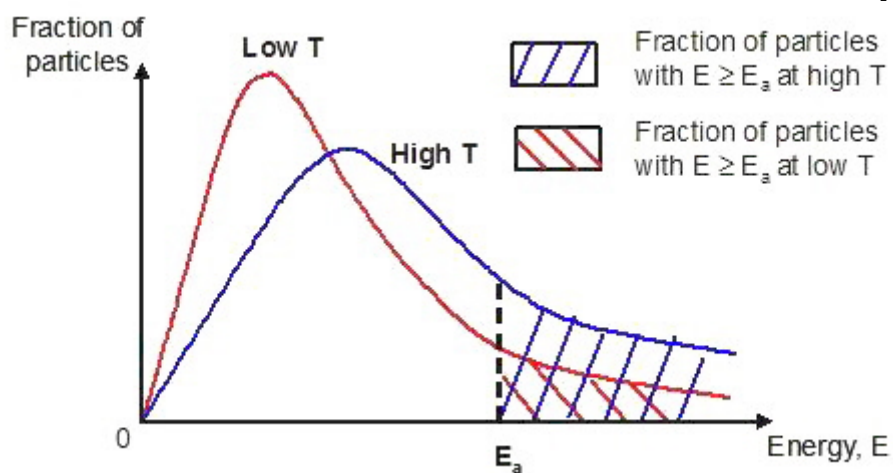
Answer:

$[(\text{CH}_3)_3\text{CBr}] / \text{mol dm}^{-3}$



- (iii) With the aid of an appropriate sketch of the Maxwell-Boltzmann distribution curve, explain why an increase in temperature can speed up this chemical reaction.

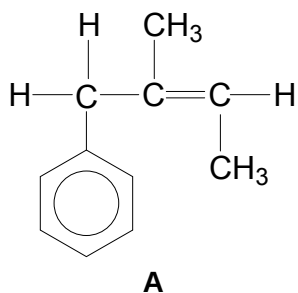
[Total:10]



Effect of increasing the temperature :

- Average **kinetic energy** of the particles **increases**.
- Frequency of **effective collisions** taking place in the reaction **increases**.
- There is an **increase** in the **fraction of particles with energy equal to or greater than the activation energy, E_a** .
- Since rate of reaction is proportional to the frequency of effective collisions, **rate of reaction increases**.

- 2 (a) Draw the structures of the major organic product(s) formed when **A** reacts with each of the following reagents.

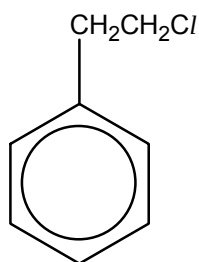
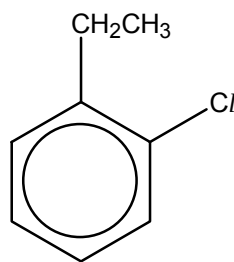


[5]

Reagents and conditions	Major organic product(s)
(i) HBr(g)	Ans
(ii) Alkaline KMnO ₄ , reflux	Ans
(iii) Cl ₂ (g) / AlCl ₃ , in the dark	Ans

- (b) Describe a simple chemical test that can be used to distinguish between **B** and **C**. State any observation(s) made and write equation(s) where appropriate.

[4]

**B****C**

Add NaOH(aq) to each compound separately and heat.

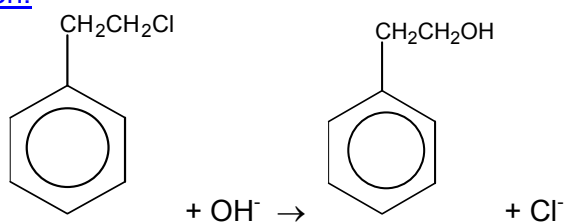
Add excess HNO₃(aq) followed by addition of AgNO₃(aq)

Observation:

B reacts to form a white ppt of AgCl.

C does not form ppt.

Equation:



[Total : 9]

3 (a) The pH of a $0.100 \text{ mol dm}^{-3}$ solution of chloroethanoic acid is 2.49.

(i) Calculate the concentration of hydrogen ions in the solution.

$$[\text{H}^+] = 10^{-2.49} = 3.24 \times 10^{-3} \text{ mol dm}^{-3}$$

(ii) Explain whether chloroethanoic acid is a strong or weak acid.

chloroethanoic acid is a weak acid because $[\text{H}^+] \ll [0.100]$, indicating it undergoes partial dissociation

(iii) The K_a expression of chloroethanoic acid is as follows

$$K_a = \frac{[\text{CH}_2\text{C}/\text{COO}^-][\text{H}^+]}{[\text{CH}_2\text{C}/\text{COOH}]}$$

Use the K_a expression given to calculate the value of K_a for chloroethanoic acid.

Ans:

$$K_a = \frac{(3.24 \times 10^{-3})^2}{0.100} = 1.05 \times 10^{-4} \text{ mol dm}^{-3}$$

(b) In an experiment, 25.0 cm^3 of $0.100 \text{ mol dm}^{-3}$ solution of chloroethanoic acid was completely neutralised by $0.100 \text{ mol dm}^{-3}$ of aqueous sodium hydroxide. [4]

(i) Calculate the number of moles and volume of sodium hydroxide needed for neutralisation.

No of moles of chloroethanoic acid = $25/1000 \times 0.100 = 2.5 \times 10^{-3} \text{ mol}$

No of moles of sodium hydroxide = $2.5 \times 10^{-3} \text{ mol}$

Volume of sodium hydroxide needed = $2.5 \times 10^{-3} / 0.100 = 25 \text{ cm}^3$

(ii) Suggest the indicator required and the appropriate colour change for this titration.

Phenolphthalein.

Colourless to Pink

(c) State and explain the acid strength of chloroethanoic acid compared with that of ethanoic acid.

Chloroethanoic acid is more acidic than ethanoic acid.

Cl atom is electron-withdrawing and disperses the negative charge on oxygen atom of $\text{CH}_2\text{ClCOO}^-$.

making $\text{CH}_2\text{ClCOO}^-$ more stable than CH_3COO^- .

[Total: 9]

- 4 (a) A college student carried out an experiment to determine the standard enthalpy change of combustion of methylacetylene, $\text{CH}_3\text{C}\equiv\text{CH}$ contained in the gas cylinder of a gas stove.

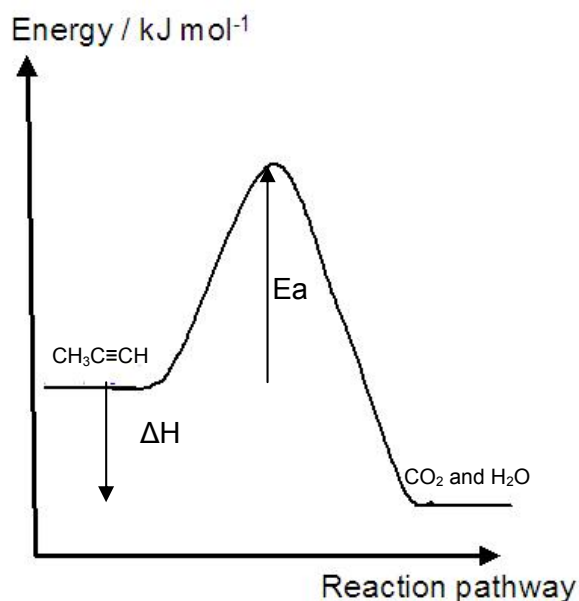
A large beaker of water was placed on the stove and heated. The initial and final temperatures were recorded. The cylinder was weighed before and after the experiment to determine the mass of methylacetylene used. This process was known to be only 70% efficient. The following data were recorded by the student.

Mass of methylacetylene before heating = 4.66 g
 Mass of methylacetylene after heating = 2.16 g
 Mass of water heated = 750 g
 Initial Temperature = 23.0 °C
 Final Temperature = 48.8 °C

- (i) Calculate the heat absorbed by the water.
 $\text{Heat absorbed} = mc\Delta T = 4.18 \times 750 \times 25.8 = 80883 = 80.9 \times 10^3 \text{ J}$
- (ii) Calculate the heat given out by methylacetylene during the experiment.
 $\text{Heat given out} = 100/70 \times 80883 = 115547 = 116 \times 10^3 \text{ J}$
- (iii) Calculate the enthalpy change of combustion of methylacetylene.
 $\text{Mass of methylacetylene} = 4.66 - 2.16 = 2.5 \text{ g}$
 $\text{Moles of methylacetylene} = 2.5/40 = 0.0625 \text{ mol}$
 $\text{Enthalpy change} = -115547/0.0625 = -1849 = -1850 \text{ kJ mol}^{-1}$
- (iv) Use the information above to construct the reaction pathway diagram for the combustion of methylacetylene. Label the E_a and ΔH clearly.

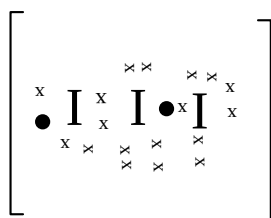
Points to look out for:

- 2 horizontal lines, one labelled reactants/ $\text{CH}_3\text{C}\equiv\text{CH}$, the other products/ CO_2 and H_2O , products lower in energy
- Two horizontal lines joined by a "hump",
- with E_a
- and ΔH (symbol or numerical value) shown



- (b) Potassium iodide dissolves in water while iodine does not. However, iodine crystals dissolve readily in a solution containing potassium iodide due to its reaction with iodide ions to form I_3^- ions.

- (i) Draw the dot-and-cross diagram for the ion, I_3^- .



- (ii) Although the I_3^- ion exists, the F_3^- ion does not. Explain why the F_3^- ion does not exist.

Fluorine atom does not have energetically accessible d-orbitals/low energy vacant 3d orbitals unlike the iodine atom.

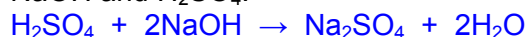
Hence, fluorine atom is unable to expand its octet configuration.

- (c) A 0.315 g sample of a Group II carbonate, ECO_3 was reacted with 25.0 cm^3 of $0.108 \text{ mol dm}^{-3}$ H_2SO_4 . The solution was boiled off to remove all the dissolved CO_2 and was then titrated with $0.124 \text{ mol dm}^{-3}$ NaOH . It was found that 17.75 cm^3 of the aqueous NaOH was required for complete neutralisation with excess H_2SO_4 .

- (i) Write an equation for the reaction taking place between the Group II carbonate, ECO_3 and aqueous sulfuric acid.



- (ii) Write an equation for the reaction taking place during the titration between NaOH and H_2SO_4 .



- (iii) Calculate the number of moles of sulfuric acid that has reacted with sodium

hydroxide for titration.

$$\begin{aligned}\text{No. of moles of NaOH used for titration} &= (17.75 / 1000) \times 0.124 \\ &= 2.201 \times 10^{-3} \text{ mol}\end{aligned}$$

$$\begin{aligned}\text{No. of moles of H}_2\text{SO}_4 \text{ needed for neutralisation} \\ &= \frac{1}{2} \times 2.201 \times 10^{-3} = 1.101 \times 10^{-3} \text{ mol} \\ &= 1.10 \times 10^{-3} \text{ mol}\end{aligned}$$

- (iv) Calculate the number of moles of sulfuric acid reacted with the metal carbonate, ECO_3 .

$$\text{No. of moles of H}_2\text{SO}_4 \text{ used initially} = (25.0/1000) \times 0.108 = 2.70 \times 10^{-3} \text{ mol}$$

$$\begin{aligned}\text{No. of moles of H}_2\text{SO}_4 \text{ reacted with ECO}_3 &= 2.70 \times 10^{-3} - 1.101 \times 10^{-3} \\ &= 1.599 \times 10^{-3} \\ &= 1.60 \times 10^{-3} \text{ mol}\end{aligned}$$

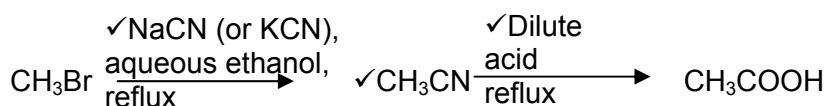
- (v) Calculate the M_r of the metal carbonate, ECO_3 .

$$\text{No. of moles of ECO}_3 \text{ present} = 1.599 \times 10^{-3}$$

$$M_r \text{ of ECO}_3 = 0.315 / (1.599 \times 10^{-3}) = 197.0$$

[Total:12]

- 5 (a) Ethanoic acid can be synthesised from bromomethane. Outline a 2-step synthesis to obtain ethanoic acid from bromomethane, stating clearly the reagents and conditions used in each synthesis. Identify the intermediate.
From bromomethane:



Step 1: NaCN in ethanol, heat with reflux.

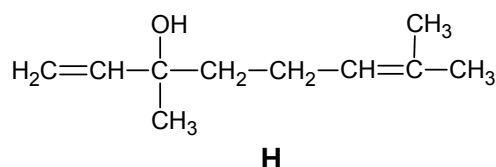
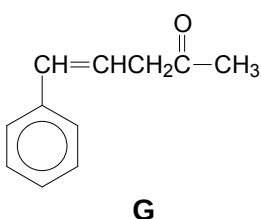
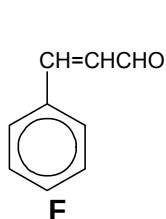
Step 2: Dilute H_2SO_4 or dilute HCl, heat with reflux.

Intermediate: CH_3CN

[3]

- (b) Recent research has suggested that cinnamon could be an effective pesticide against the larvae of mosquitoes, thus helping in the fight against malaria.

Each of the following three compounds, which are present in cinnamon, appears to be effective as a pesticide.



For each of the compounds **F** to **H**, state the reagents and conditions which would distinguish it from the other two and describe the observations that would be seen.

Compound F:

Test: Add Tollens' reagent (to the 3 samples) separately and warm.
 Observation: Silver mirror formed.

Alt. test: Fehling's reagent; warm

Compound G:

Test: aqueous iodine, NaOH (aq), warm.

Observation: Yellow ppt of CHI₃ was observed, orange-brown iodine decolourised

Compound H:

Test: Add Na solid at room temperature

Observation: effervescence of H₂ gas with pop' sound with lighted splint.

Alt. test: (i) SOCl₂; heat (ii) PCl₅, room temperature (iii) PCl₃, room temperature.

Note: (to the 3 samples) may not be written explicitly if student imply that in the answers.

[6]

- (c) When compound **J** with a molecular formula of C₁₀H₁₂ is treated with cold dilute alkaline potassium manganate(VII), **J** forms compound **K**.

However, on heating **J** with acidified potassium manganate(VII), 2 products, compound **L**, C₈H₈O and compound **M** are formed. **M** has the following composition by mass: C, 40.0%; H, 6.7%; O, 53.3%.

On refluxing **K** with ethanoic acid in the presence of concentrated sulfuric acid, compound **N**, C₁₄H₁₈O₄ is obtained.

- (i) Calculate the empirical formula of compound **M** and hence, determine its molecular formula, given the *M_r* of compound **M** is 60.0.

	C	H	O
% mass(in 100g)	40.0	6.7	53.3
Amount in mol	3.333	6.7	3.331
Divide by smallest mol	1	2.011	1
Simplest ratio	1	2	1

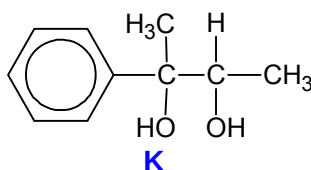
Empirical formula of **M** is CH₂O

Let the molecular formula of **S** be (CH₂O)_{*n*}

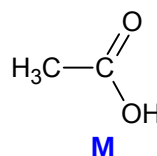
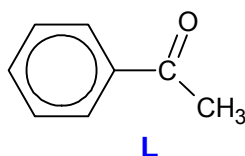
$n[12.0 + 2(1.0) + 16.0] = 60.0 \rightarrow n = 2$

Molecular formula of **M** is C₂H₄O₂

- (ii) Draw the structures of **K**, **L** and **M** and explain all the reactions above.



10



J undergoes addition to form K, a diol.

J undergoes oxidative cleavage to form L, a ketone and M, carboxylic acid.

K undergoes esterification to form N, an ester.

[8]

- (d) Explain, in terms of structure and bonding, why the melting points of Na₂O and MgO are higher than 900 °C but the melting point of P₄O₆ is less than 500 °C.

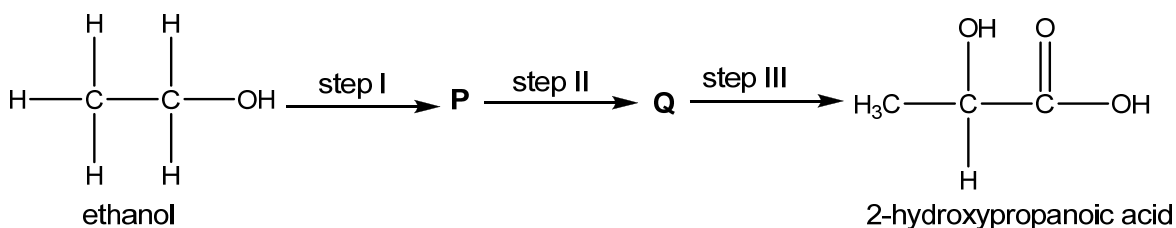
Na₂O, MgO form giant ionic lattice, large amount of energy required to break strong electrostatic attractions between oppositely charged ions (ionic bonds between ions), hence they have high melting points.

P₄O₆ exist as simple covalent molecules / simple molecular substance, lesser energy required to break the weak Van der Waals forces between molecules, hence it has a low m.p.

[3]

[Total:20]

- 6 The reaction scheme below shows the synthesis of 2-hydroxypropanoic acid from ethanol.



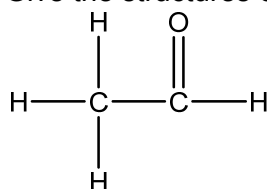
- (a) (i) State the reagents and conditions used for step I, II and III.

Step 1: K₂Cr₂O₇, dil H₂SO₄, heat with distillation

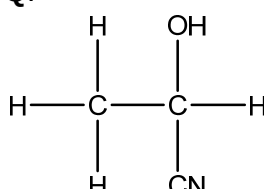
Step 2: HCN, trace amount of NaOH OR NaCN, 10-20°C

Step 3: dilute HCl, or dil H₂SO₄, heat

- (ii) Give the structures of P and Q.



P



Q

- (iii) Describe a simple chemical test to distinguish between propanoic acid and 2-hydroxypropanoic acid.

KMnO₄, dil H₂SO₄, heat

2-hydroxypropanoic acid: Purple KMnO_4 decolourises

propanoic acid: Purple KMnO_4 remains

OR

$\text{K}_2\text{Cr}_2\text{O}_7$, dil H_2SO_4 , heat

2-hydroxypropanoic acid: Orange KCr_2O_7 decolourises

propanoic acid: Orange KCr_2O_7 remains

OR

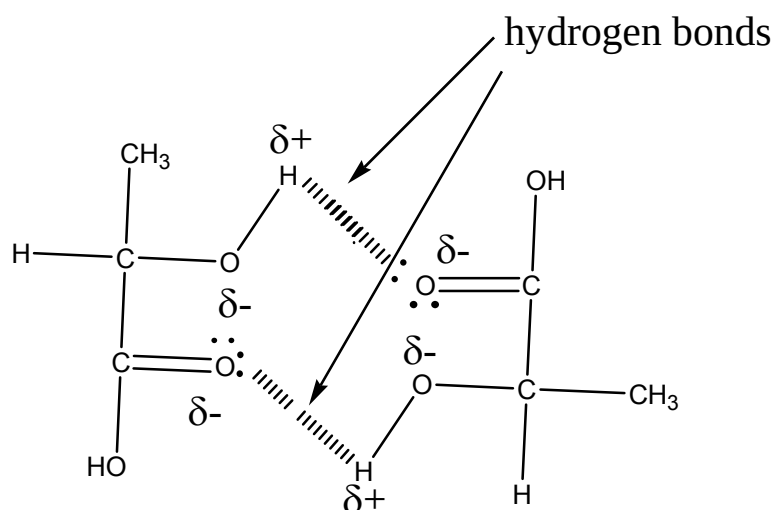
Aq I_2 , Aq NaOH , warm

2-hydroxypropanoic acid: yellow ppt of CHI_3 observed

propanoic acid: no yellow ppt if CHI_3 observed

- (iv) 2-hydroxypropanoic acid dimerises in an organic solvent such as tetrachloromethane. Draw a clearly labelled diagram of the dimer.

[9]

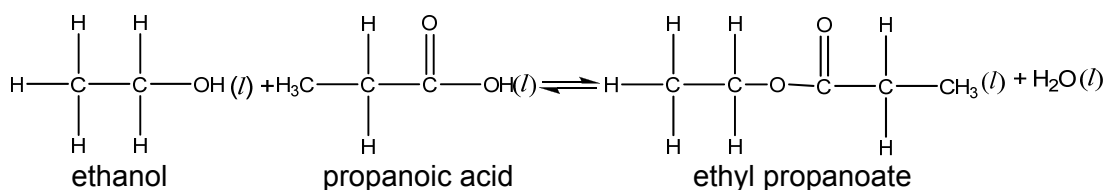


Polarity of OH bond and lone pair of electron on oxygen

Dash line between H and O and Labeling of H-bonding

Show 2 hydrogen bonds

- (b) Equal amount of ethanol and propanoic acid were mixed and allowed to reach equilibrium at temperature T . One-third of the acid was found to be in the equilibrium mixture.



- (i) Calculate the equilibrium constant, K_c , for the reaction at temperature T .

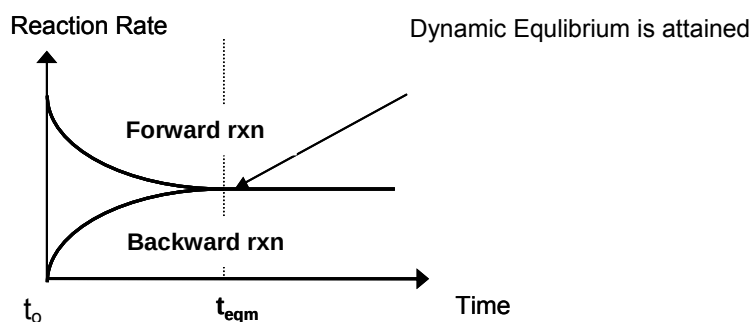
	$ \begin{array}{c} \text{H} \quad \text{H} \\ \quad \\ \text{H}-\text{C}-\text{C}-\text{OH} \quad (l) \\ \quad \\ \text{H} \quad \text{H} \\ \text{ethanol} \end{array} + \begin{array}{c} \text{H} \quad \text{O} \\ \quad \\ \text{H}_3\text{C}-\text{C}-\text{C}-\text{OH} \quad (l) \\ \\ \text{H} \\ \text{propanoic acid} \end{array} \rightleftharpoons \begin{array}{c} \text{H} \quad \text{H} \quad \text{O} \quad \text{H} \\ \quad \quad \quad \\ \text{H}-\text{C}-\text{C}-\text{O}-\text{C}-\text{C}-\text{CH}_3 \quad (l) \\ \quad \quad \quad \\ \text{H} \quad \text{H} \quad \text{H} \quad \text{H} \\ \text{ethyl propanoate} \end{array} + \text{H}_2\text{O} \quad (l) $			
Initial no. of moles	x	x	0	0
Change in moles	-2/3x	-2/3x	+2/3x	+2/3x
No. of moles at equilibrium	1/3x	1/3 x	2/3x	2/3x

$$K_c = \frac{\left(\frac{2}{3}x\right)^2}{\frac{1}{\left(\frac{3}{V}\right)^2}} = 4 \quad \text{V cancels out}$$

- (ii) How will the yield of ester formed be affected if anhydrous calcium chloride is added to the mixture?

The anhydrous calcium chloride will absorb the water. By LCP, the equilibrium will shift to the right to produce more water. As a result, more ester will be formed and the yield will increase.

- (iii) Sketch a rate-against-time graph for the forward and reverse reactions from t_0 , time when the reagents are mixed, to t_{eqm} , time when the reaction reaches equilibrium. Label the two lines clearly.



- (c) In order to study the amount of acid left in the equilibrium mixture, the equilibrium mixture was titrated rapidly with 0.5 mol dm^{-3} NaOH. Given that 25.0 cm^3 of the equilibrium mixture from (b) required 31.50 cm^3 of NaOH for complete neutralisation, what is the original concentration of propanoic acid amount in a 25.0 cm^3 sample? [7]

No. of moles of NaOH = $0.5/1000 \times 31.50 = 0.01575$

No. of moles of propanoic acid at equilibrium = 0.01575

Original No. of moles of propanoic acid = $0.01575 \times 3 = 0.04725 \text{ moles}$

Concentration of propanoic acid = $0.04725/25 \times 1000 = 1.89 \text{ mol dm}^{-3}$

- (d) Some data for eight consecutive elements, represented by R to Y are given below. [2]

Element	R	S	T	U	V	W	X	Y
Atomic Number	n	n+1	n+2	n+3	n+4	n+5	n+6	n+7
1 st Ionisation energy/ kJ mol^{-1}	1090	1400	1310	1680	2080	494	736	577

Given that the elements do not belong to the same period and using the information above, state the Group numbers of element V and W.

V is in group 0/VIII and W is in group I

[2]

- 7 Chromates have been traditionally used in chrome plating to protect metals from corrosion and to improve paint adhesion. However, due to toxicity issues, its uses have become more limited. One such compound is sodium chromate(VI), $\text{Na}_2\text{Cr}_2\text{O}_7$, a green solid.

When mixed with dilute H_2SO_4 , it disproportionates to yield $\text{Cr}^{3+}(\text{aq})$ and $\text{Cr}_2\text{O}_7^{2-}(\text{aq})$. The mixture produced was then mixed with excess KI. The I_2 liberated required 40.0cm^3 of $0.25\text{ mol dm}^{-3}\text{ Na}_2\text{S}_2\text{O}_3(\text{aq})$ for reaction.

The 3 reactions described are as follows:

Reaction 1: Disproportionation of CrO_4^{3-}

Reaction 2: $\text{Cr}_2\text{O}_7^{2-}(\text{aq}) + 6\text{I}^-(\text{aq}) + 14\text{H}^+(\text{aq}) \rightarrow 2\text{Cr}^{3+}(\text{aq}) + 3\text{I}_2(\text{aq}) + 7\text{H}_2\text{O}(\text{l})$

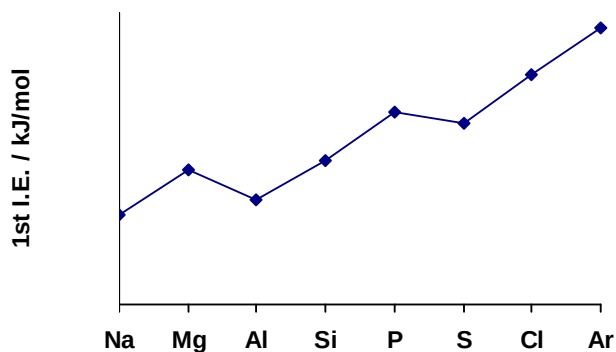
Reaction 3: $\text{I}_2(\text{aq}) + 2\text{S}_2\text{O}_3^{2-}(\text{aq}) \rightarrow 2\text{I}^-(\text{aq}) + \text{S}_4\text{O}_6^{2-}(\text{aq})$

- (a) (i) Write the 2 balanced half equations occurring in Reaction 1 in an acidic medium:
- I) CrO_4^{3-} is reduced to Cr^{3+}
 - II) CrO_4^{3-} is oxidised to $\text{Cr}_2\text{O}_7^{2-}$
- I) $\text{CrO}_4^{3-} + 8\text{H}^+ + 2\text{e}^- \rightarrow \text{Cr}^{3+} + 4\text{H}_2\text{O}$
- II) $2\text{CrO}_4^{3-} + 2\text{H}^+ \rightarrow \text{Cr}_2\text{O}_7^{2-} + \text{H}_2\text{O} + 2\text{e}^-$
- (ii) Use the half-equations you have written in a(i) to write the overall equation with state symbols for **Reaction 1**.
- $3\text{CrO}_4^{3-}(\text{aq}) + 10\text{H}^+(\text{aq}) \rightarrow \text{Cr}^{3+}(\text{aq}) + \text{Cr}_2\text{O}_7^{2-}(\text{aq}) + 5\text{H}_2\text{O}(\text{l})$
- (iii) You are given that 1 mole of $\text{Cr}_2\text{O}_7^{2-}$ liberates iodine which requires 6 moles of $\text{S}_2\text{O}_3^{2-}$ for reaction. Calculate the amount of $\text{Cr}_2\text{O}_7^{2-}$ that is formed from the disproportionation reaction.
- moles of $\text{S}_2\text{O}_3^{2-} = 0.04 \times 0.25 = 0.01\text{ mol}$
- moles of $\text{Cr}_2\text{O}_7^{2-} = 0.01/6 = 0.00167\text{ mol}$
- (iv) Hence, calculate the mass of $\text{Na}_2\text{Cr}_2\text{O}_7$ used in this reaction.
- moles of $\text{CrO}_4^{3-} = 3 \times 0.00167 = 0.00501\text{ mol}$
- mass of $\text{Na}_2\text{Cr}_2\text{O}_7 = 0.00501 \times 185 = \underline{0.927\text{g}}$

[7]

(b) Besides transition elements, the main group elements and their oxides are also well-studied.

(i) Sketch the general trend of the first ionisation energy of the elements across the third period of the Periodic Table. Explain the general shape of your graph.



- Across the period, nuclear charge increases due to increasing proton number.
- Shielding effect remains relatively constant across the period as electrons are added to the same quantum shell.
- Hence effective nuclear charge increases. **OR** This results in a stronger attraction between valence electrons and nucleus
- More energy is required for removal of valence electron

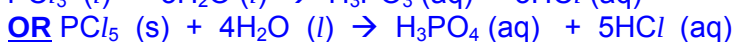
(ii) When the chlorides of Mg and P are dissolved in water, the pH of the resulting solutions are 6.5 and 2 respectively. Explain these reactions and write equations with state symbols where appropriate.

MgCl_2 first undergoes hydration

$\text{MgCl}_2 \rightarrow \text{Mg}^{2+}(\text{aq}) + 2\text{Cl}^{-}(\text{aq})$. However, Mg^{2+} ions have high charge density and undergo hydrolysis to produce H^{+} ions.

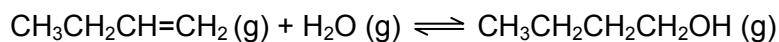


PCl_3 and/OR PCl_5 undergo hydrolysis

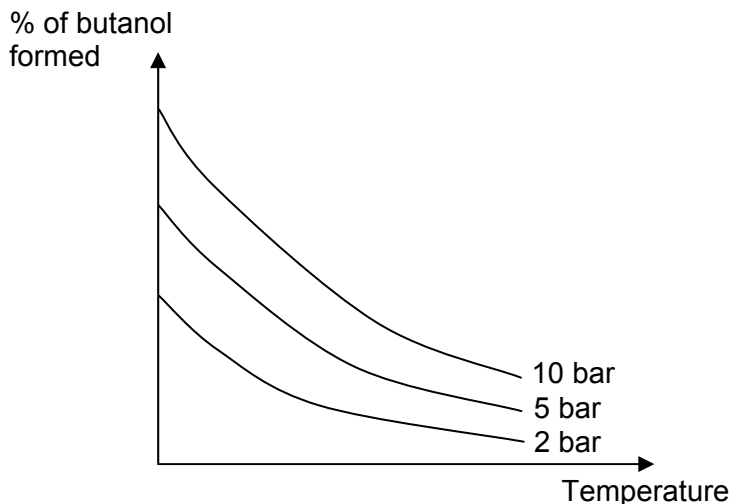


[6]

- (c) The hydration of gaseous butene to gaseous butanol is represented by the following equilibrium.



The percentage of butanol formed at equilibrium at various temperatures and pressures is shown below.



With reference to the graph above, explain the trend in the percentage of butanol formed with changes in

- (i) Temperature

As temperature increases, the percentage of butanol formed decreases.

This shows that equilibrium position has shifted to the left to absorb additional heat, ~~fav~~, favouring the endothermic reaction.

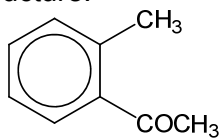
- (ii) Pressure

As pressure increases, the percentage of butanol formed increases.

This shows that equilibrium position has shifted to the right to produce less gaseous molecules so as to decrease the pressure.

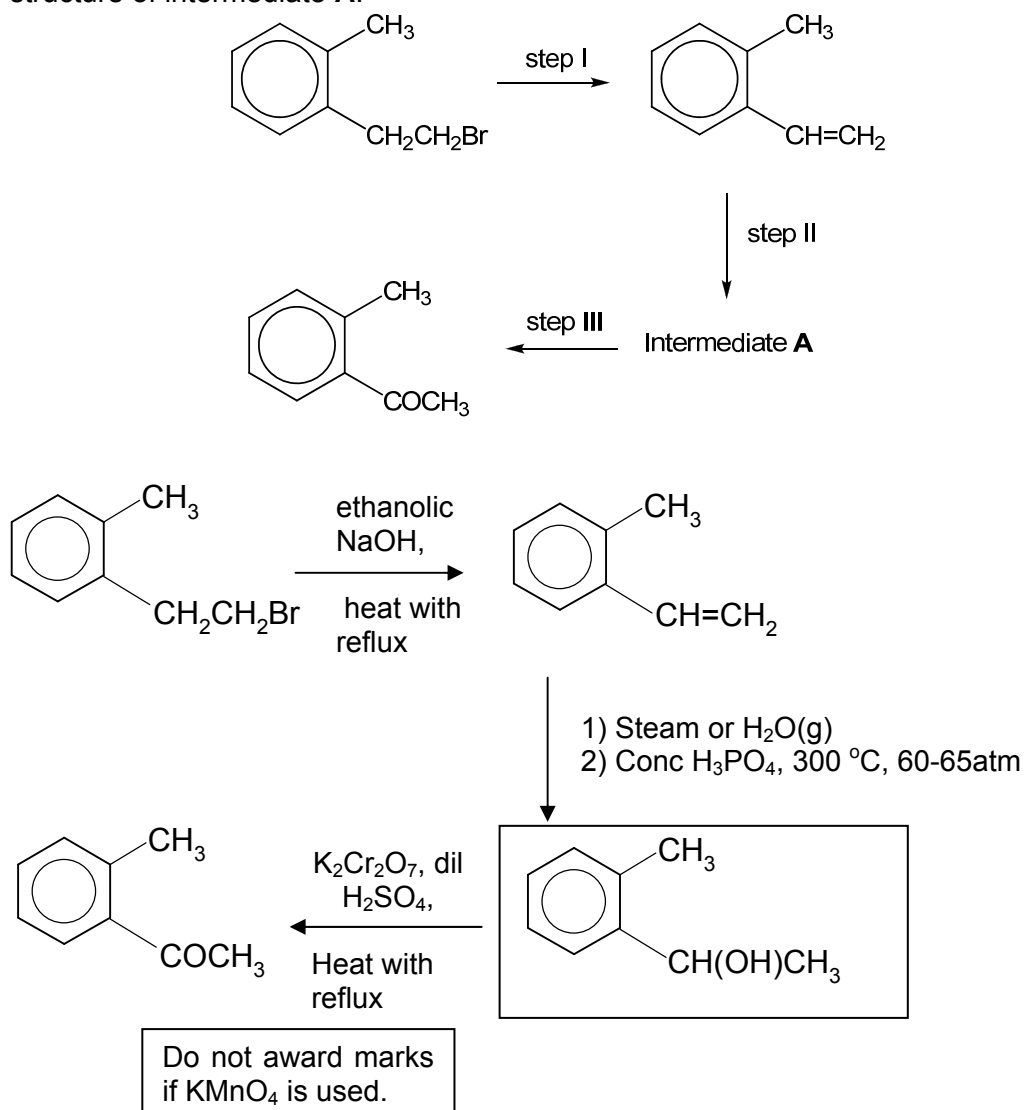
[3]

- (d) Compound Z has the following structure:



Z

The following synthetic route shows how compound **Z** can be obtained in a three-step synthesis. State the reagents and conditions for all steps and draw the structure of intermediate **A**.



Alternative answer for Electrophilic addition of water:

- 1) conc. H_2SO_4
- 2) Steam or $\text{H}_2\text{O(g)}$

[4]

[Total: 20]